Energy and AI

Energy demand from AI

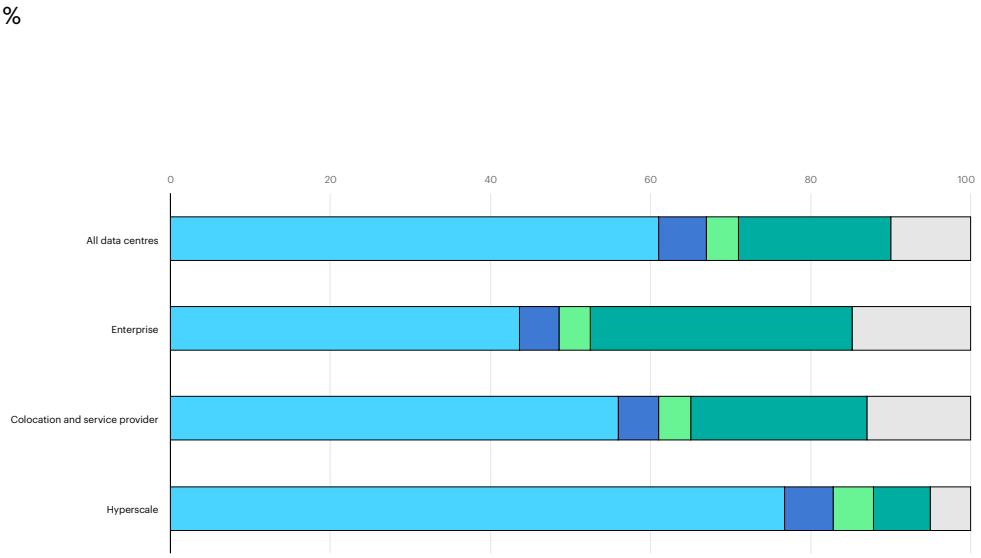
What is a data centre?

Artificial intelligence (AI) model training and deployment occur mainly in data centres. Understanding the role of data centres as actors in the energy system first requires an understanding of their component parts. Data centres are facilities used to house servers, storage systems, networking equipment and associated components that are installed in racks and organised into rows. This IT equipment, and a range of auxiliary equipment required to keep it in working order, comprise the following:

- **Servers** are computers that process and store data. They can be equipped with central processing units (CPUs) and specialised accelerators such as graphics processing units (GPUs). On average they account for around 60% of electricity demand in modern data centres, although this varies greatly between data centre types.

Share of electricity consumption by data centre and equipment type, 2024

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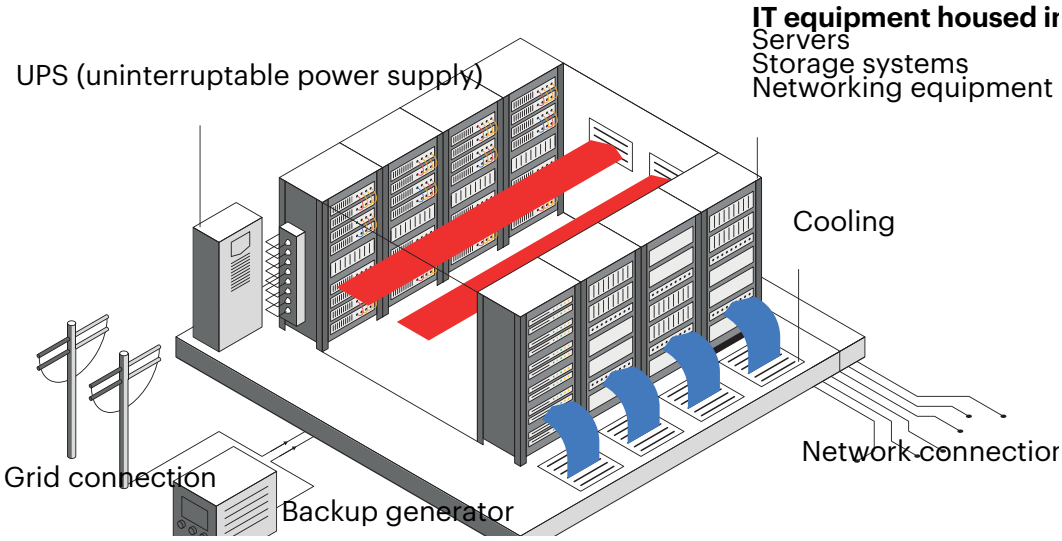


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Servers Storage Network Cooling Other infrastructure

- **Storage systems** are devices used for centralised data storage and backup, and account for around 5% of electricity consumption.
- **Networking equipment** include switches to connect devices within the data centre, routers to direct traffic and load balancers to optimise performance. Networking equipment accounts for up to 5% of electricity demand.
- **Cooling and environmental control** refers to equipment that regulates temperature and humidity to keep IT equipment operating at optimal conditions. The share of cooling systems in total data centre consumption varies from about 7% for efficient hyperscale data centres to over 30% for less-efficient enterprise data centres.
- **Uninterruptible power supply (UPS) batteries and backup power generators** are there to keep the data centre powered during outages. Both UPS and backup generators are rarely used, but necessary to ensure the extremely high levels of reliability that data centres must meet.
- **Other infrastructure**, such as lighting and office equipment for on-site staff, etc.

The share of these different components in data centre electricity consumption varies greatly by type of data centre, depending on the nature and efficiency of the equipment they have installed.


IT equipment housed ir
Servers
Storage systems
Networking equipment
UPS (uninterruptable power supply)
Cooling
Network connectior
Grid connection
Backup generator

The outlook for energy demand from data centres

Data centres – at least at the scale seen today – are relatively new actors in the energy system at the global level. Today, electricity consumption from data centres is estimated to amount to around 415 terawatt hours (TWh), or about 1.5% of global electricity consumption in 2024. It has grown at 12% per year over the last five years.

The rise of AI is accelerating the deployment of high-performance accelerated servers, leading to greater power density in data centres. Understanding the pace and scale of accelerator adoption is critical, as it will be a key determinant of future electricity demand. The key input to our modelling is therefore near-term industry projections for server shipments, considering the outlook for demand and supply constraints.

There is substantial uncertainty both about data centre consumption today and in the future. The uncertainty surrounding future electricity demand requires a scenario-based approach to explore alternative pathways and provide perspectives on timelines relevant for energy sector decision-making. While the technology sector moves quickly and a data centre can be operational in two to three years, the broader energy system requires longer lead times to schedule and build infrastructure, which often requires extensive planning, long build times and high upfront investment.

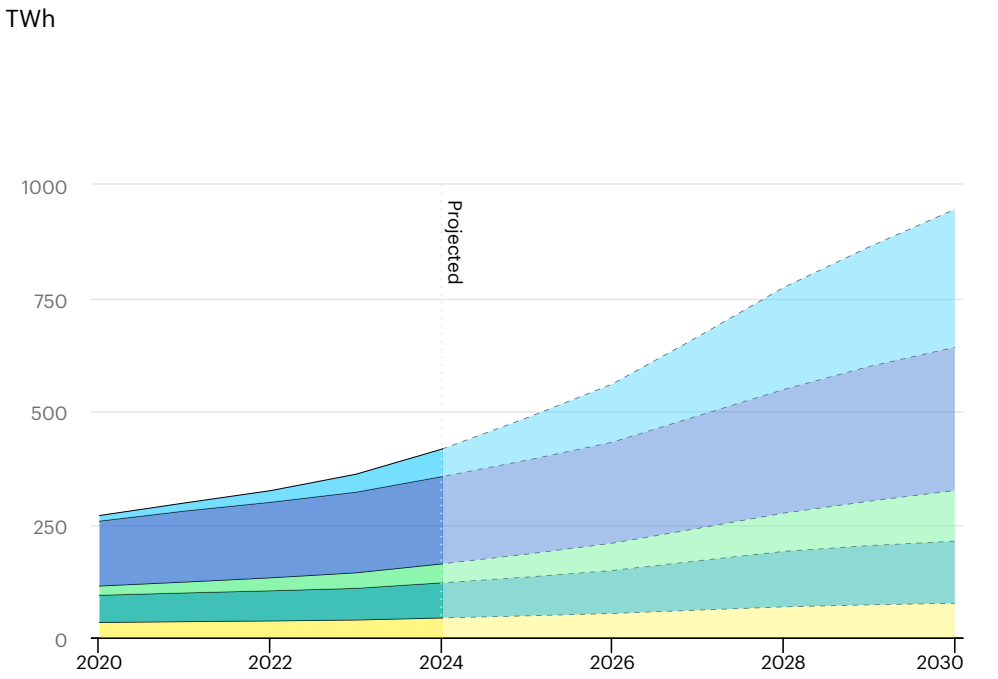
Three sensitivity cases (Lift-Off, High Efficiency and Headwinds) capture uncertainties in efficiency improvements in hardware and software, AI uptake and energy sector bottlenecks.

Our Base Case finds that global electricity consumption for data centres is projected to double to reach around 945 TWh by 2030 in the Base Case, representing just under 3% of total global electricity consumption in 2030. From 2024 to 2030, data centre electricity consumption grows by around 15% per year, more than four times faster than the growth of total electricity consumption from all other sectors. However, in the wider context, a 3% share in 2030 means that data centre share in global electricity demand remains limited.

Electricity consumption in accelerated servers, which is mainly driven by AI adoption, is projected to grow by 30% annually in the Base Case, while conventional server electricity consumption growth is slower at9% per year. Accelerated servers account for almost half of the net increase in global data centre electricity consumption, while conventional servers account for only around 20%. Other IT equipment, and infrastructure (cooling and other infrastructure) demand account for around 10% and 20% of the net increase respectively. All three types of data centres – enterprise, colocation and server provider, and hyperscale – contribute to the growth in electricity consumption.

Global data centre electricity consumption, by equipment, Base Case, 2020-2030

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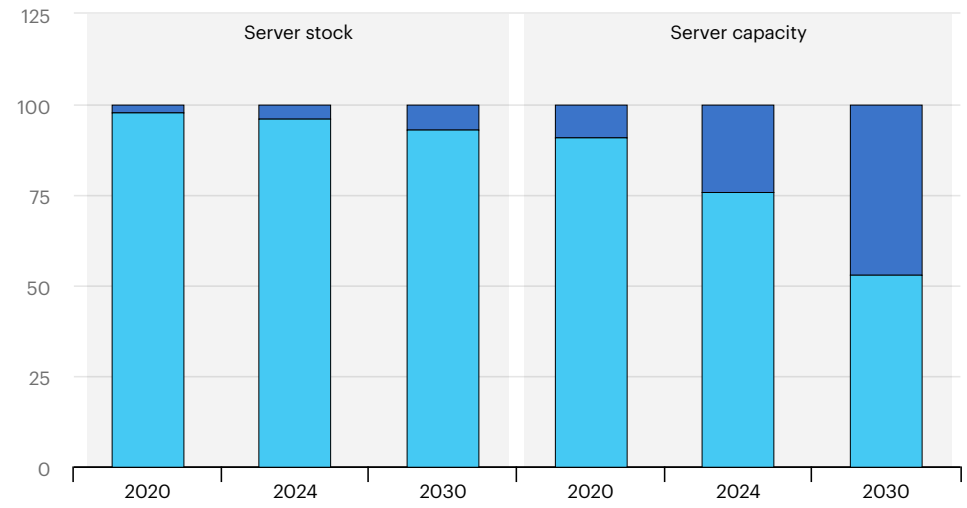


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Accelerated servers · Conventional servers · Other IT equipment · Cooling · Other infrastructure

Global data centre electricity consumption in Server stock and in Server capacity, 2020-2030

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Conventional servers Accelerated servers

Regional outlook

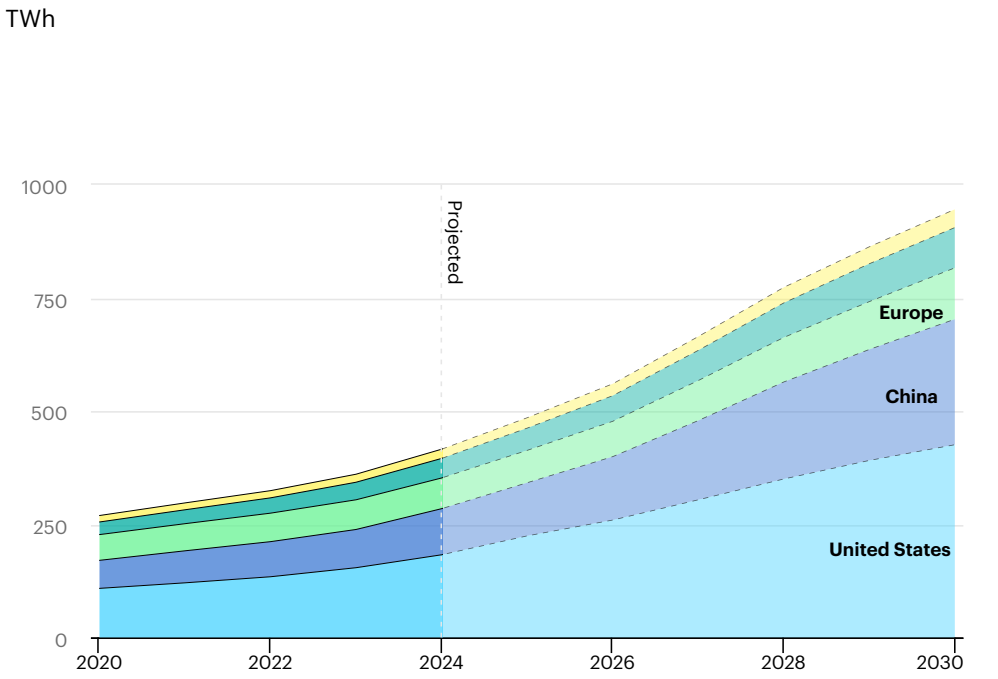
The United States, China and Europe are projected to remain the largest regions for data centre electricity demand over the coming years. However, other regions are experiencing strong growth in data centre development, positioning them to play increasingly important roles in the global data centre landscape. A notable example is Southeast Asia, where electricity demand from data centres is expected to more than double by 2030, partially due to the presence of a regional hub in Singapore and southern Malaysia.

China and the United States are the most significant regions for data centre electricity consumption growth, accounting for nearly 80% of global growth to 2030. Consumption increases by around 240 TWh (up 130%) in the United States, compared to the 2024 level. In China it increases by around 175 TWh (up 170%). In Europe it grows by more than 45 TWh (up 70%). Japan increases by around 15 TWh (up 80%).

Comparing data centre electricity consumption normalised per capita can give a sense of the importance of this sector in different economies. Africa has the lowest consumption at less than 1 kWh of data centre electricity consumption per capita in 2024, rising to slightly less than 2 kWh per capita by the end of the decade. However, there are strong differences within the region, with South Africa showing strong growth and per-capita consumption more than 15 times larger than the continental average in 2030, with an intensity higher than 25 kWh per capita. By contrast, the United States has the highest per-capita data centre consumption, at around 540 kWh in 2024. This is projected to grow to over 1 200 kWh per capita by the end of the decade, which is roughly as much as 10% of the annual electricity consumption of an American household. This intensity is also one order of magnitude higher than any other region in the world.

Data centre electricity consumption by region, Base Case, 2020-2030

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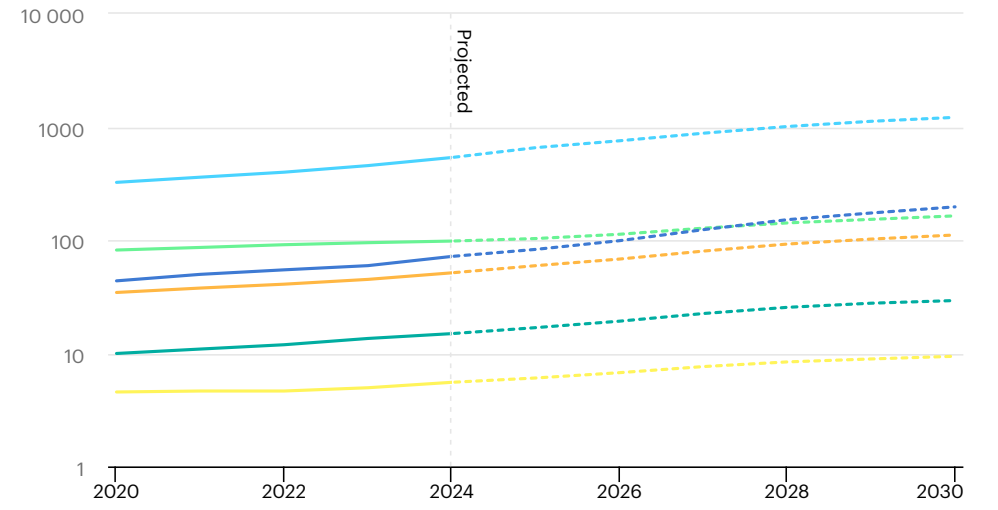


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United States China Europe Asia excluding China
Rest of world

Data centre electricity consumption per capita by region, Base Case, 2020-2030

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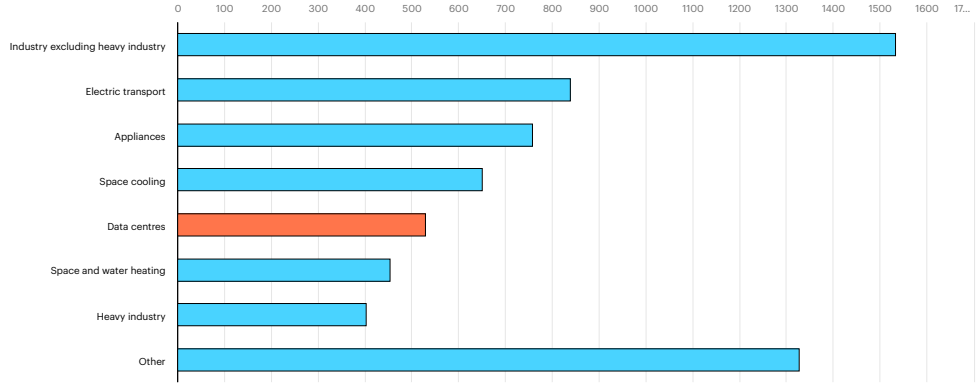
United States China Europe Asia excl. China
Rest of world Global average

Despite the strong increase, data centre electricity demand growth accounts for less than 10% of global electricity demand growth between 2024 and 2030 in the Base Case. Other key drivers, such as industry output growth and electrification, the deployment of electric vehicles, and the adoption of air conditioning, lead the way. However, while the absolute growth may appear smaller, data centres, unlike electric vehicles, tend to concentrate in specific locations, making their integration into the grid potentially more challenging.

Increase in electricity demand by sector, Base Case, 2024-2030

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TWh



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Electricity demand in the sensitivity cases

The **Lift-Off Case** assumes stronger growth in AI adoption than in the Base Case. A more resilient supply chain and greater flexibility, in data centre location, powering and operations, enable faster data centre deployment. It sees global electricity demand from data centres in 2035 that is around 45% higher than in the Base Case, exceeding the 1 700 TWh mark and reaching around 4.4% of global electricity demand.

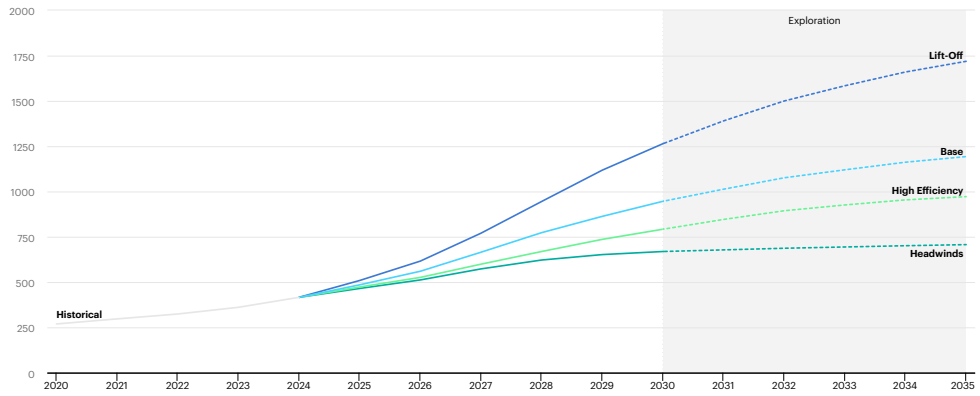
The **High Efficiency Case** shares similar constraints and drivers with the Base Case, but assumes stronger progress on energy efficiency in software, hardware and infrastructure. As a result, the same level of demand for digital services and AI is met with a reduced electricity consumption footprint. This unlocks energy savings of more than 15%, with global electricity demand from data centres reaching around 970 TWh by 2035. As a result, 2.6% of global electricity demand goes to data centres.

The **Headwinds Case** captures the impact of a downside in the outlook for data centre deployment, particularly due to slower than expected AI adoption. The emergence of local bottlenecks, along with a tight supply chain, causes delays in capacity expansion compared to the most ambitious industry projections. In this Case, the total installed IT stock by the end of the decade is projected to be smaller than in the Base Case, with growth plateauing beyond 2030 (this still means growing service demand, as the stock of IT equipment becomes more efficient over time). Similar to trends seen in the early 2010s, the improvements in efficiency are expected to offset most of the impact of increased IT stock utilisation. This leads to a plateau in energy demand at around 700 TWh, limiting the growth of the data centre share of global electricity demand to less than 2% in 2035.

Global data centre electricity consumption by sensitivity case, 2020-2035

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TWh



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Historical Base Lift-Off High Efficiency Headwinds

Next

Energy supply for AI *(/reports/energy-and-ai/energy-supply-for-ai#abstract)*